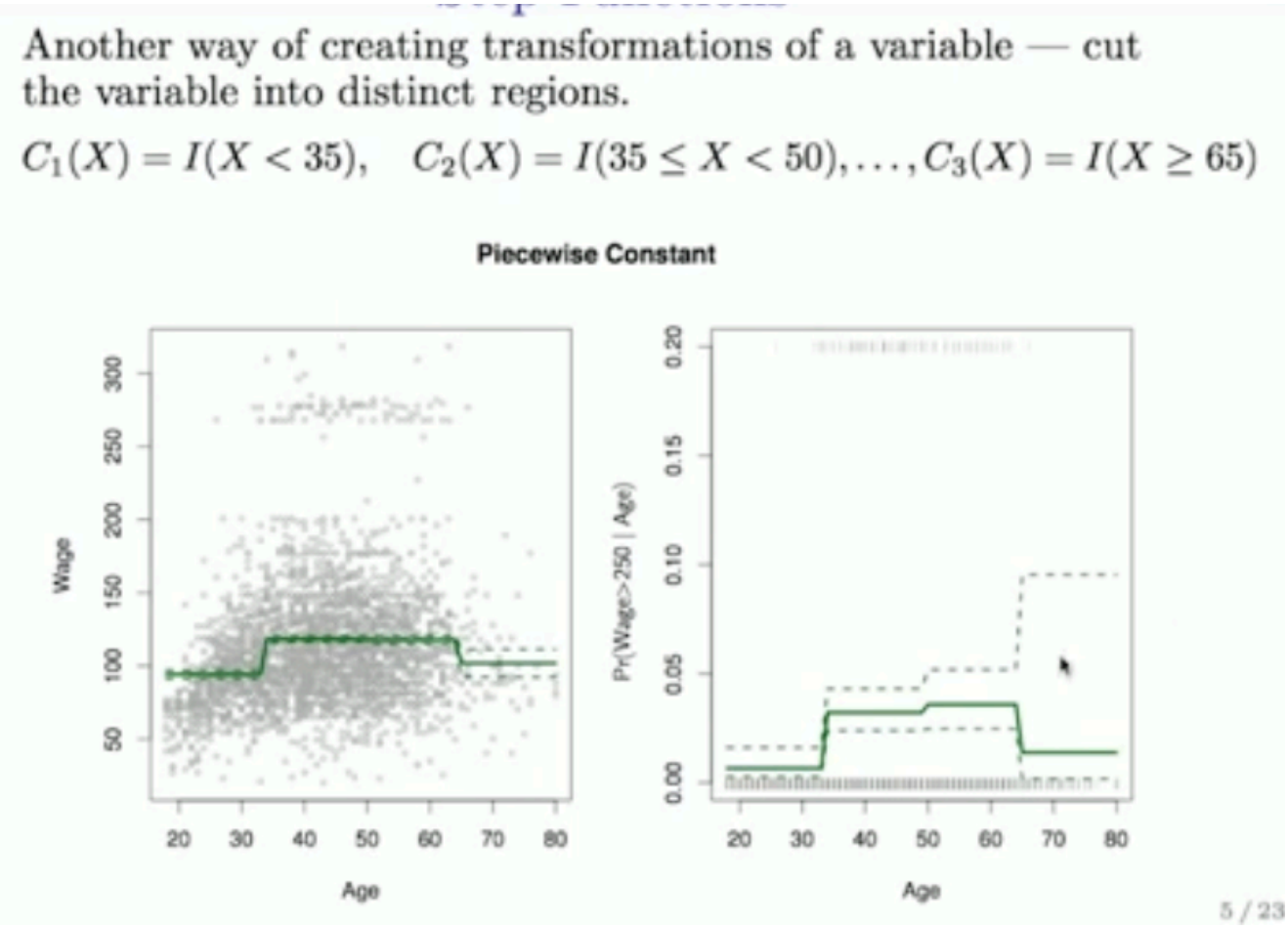
Step Function

- it uses the polynomial functions of the features as predictors in a linear model
- We use it to avoid imposing such a global structure

  Idea: Divide the range of X into several **bins** (intervals), and fit a **constant** value in each bin.
- Example: For age, create bins like <30, 30–40, 40–50, >50, and estimate a separate average wage in each bin.
- Result: Piecewise constant function --> looks like steps/jumps.
- **Very simple** but often too coarse (big jumps instead of smooth change).



- Useful way of creating interactions that are easy to interpret.
  - for example, interaction effect of year and Age:

$$I(\texttt{Year} < 2005) \cdot \texttt{Age}, \quad I(\texttt{Year} \geq 2005) \cdot \texttt{Age}$$

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